

ADR-V6-001

Architecture Decision Record

Football Primitive Registry + End-to-End WorldProjection → EntryContract → WorldLineupOutcome

With multi-sport appearance-mechanics contract, block-length selection policy, and Higher/Lower derivation law. Canonical v5.4.1 untouched. PHASE_BC_SCHEMA_V1_2026-08-25 frozen.

Document ID	DCM-ADR-V6-001-2026-08-27
Decision class	IMPLEMENT NOW (football registry + E2E settlement wiring)
Schema impact	ZERO — hash must remain 6e78dacc1984338643bdcabc7477fd3ce2dd065da1e9629646dacc21cdb1f22
Baseline	Canonical DCM v5.4.1 + accepted basketball / Phase B-C development layers
Rule snapshot	PRIZEPICKS_PLAYER_PICKS_2026-08-25_V1
Learning revision	LR000000 — architecture is not a learning promotion
Predictive claim	NONE. Engineering completeness ≠ forecast superiority.
Compiled	2026-08-27

Classification summary

Workstream	Class	Authority	Why
Football (NFL) primitive registry + conservation harness	IMPLEMENT NOW	Development baseline only	Master Blueprint §12.1 step 3. Schema already proven on basketball.
E2E WorldProjection → EntryContract → WorldLineupOutcome	IMPLEMENT NOW	Development baseline only	Blueprint §12.1 step 4 / §14.4. Wire existing adapter; do not invent payout tables.
First E2E fixture on basketball worlds, then NFL	IMPLEMENT NOW	Development baseline	Settlement must be sport-agnostic. Basketball worlds already exist.
NFL Reboot / K-P / defense exclusion rows	IMPLEMENT NOW	Fail-closed snapshot V1	Already specified. Unknown stat key → UNRESOLVED.
CFB registry	IMPLEMENT NOW only if player_reboot_eligible list is frozen	Fail-closed otherwise	Blueprint forbids broad CFB phase assumptions.
WNBA / NBA appearance + settlement reuse	KEEP / EXTEND	Already LIVE for basketball primitives	Do not rewrite basketball. Reuse worlds through new E2E path.
MLB plate-appearance engine	RESEARCH FIRST then next registry	No production authority in this ADR	Hardest discrete-state sport after football. Design now; code after football E2E is green.
Block-length selection / BEL coverage tournament	RESEARCH FIRST	Phase A roles already frozen	Politis-White is a candidate generator, not a theorem. No production BEL promotion.
AFL, CFL, LAX, handball, cricket, golf, tennis, soccer/EPL	DEFER registry code	Schema-ready, not production-ready	SportFamily plugin later. Do not mint new ledger types.
NFL preseason (NFLP)	FAIL CLOSED	Snapshot V1: no preseason reboot inference	May model physically. May not guess PrizePicks preseason settlement.
OTD / unnamed boards	FAIL CLOSED	Unknown platform rule	A code name is not a verified MarketDefinition.
Tracking toys (NGS, SportVU, Statcast, Hawk-Eye) as direct prop generators	REJECT as production inputs	Evidence layer only	Coordinates are evidence, not primitives, until a versioned definition maps them.
DPMM / GP-LVM / GPDM	UNCHANGED	SHADOW / RESEARCH_ONLY / DEFERRED	This ADR does not reopen those gates.
Six-pick forcing / Goblin selection / payout-from-card-size	REJECT	Inherited production law	Non-negotiable.

1. Problem

DCM can already simulate a basketball event as one primitive world and derive related markets without independently sampling composites. It cannot yet do the same for football, and it cannot yet take a frozen world projection through an exact PrizePicks EntryContract to a WorldLineupOutcome under the 2026 Player Picks economics (Minimum Guarantee, Leaderboard max-not-sum, Reboot/DNP/Tie as separate dimensions).

That gap is operational, not philosophical. The frozen Phase B/C schema already names the objects. Basketball already occupies them. PrizePicks rule tables already exist as a development module. Football is the first sport that will prove the schema is universal rather than basketball-shaped. The E2E path is the first object chain that will prove sports prediction and platform economics can stay uncontaminated while still producing a card-level return distribution.

A second, larger request sits behind that slice: make every major sport — NFL, CFB, AFL, CFL, NBA, WNBA, MLB, soccer/EPL, tennis, golf, cricket, lacrosse, handball, preseason football — produce a Higher/Lower conclusion from metrics and tracking data. That request is valid as a horizon. It is invalid as a single implementation ticket. The correct move is to freeze a sport-plugin contract now so later registries cannot invent new world/ledger/settlement types, and to refuse any sport whose platform rules or appearance mechanics are unverified.

2. Current limitation

2.1 What is already solved

- Canonical v5.4.1 event-first, unit-safe, versioned-definition law. Untouched.
- PHASE_BC_SCHEMA_V1_2026-08-25 deeply immutable contracts, including EventWorld, PrimitiveStatLedger, ConservationRule, MarketDefinition, EntryContract, WorldLineupOutcome.
- Basketball primitive topology LIVE: Minutes, FGA/3PA/2PA, makes, FTA/FTM, OREB/DREB/REB, AST, STL, BLK, TO, PTS, plus derived PRA/PR/PA/RA/Stocks/Fantasy.
- Exact basketball identities hold world-by-world, not in expectation.
- Opportunity and efficiency are already separate objects in the frozen schema.
- PrizePicks snapshot V1 and Leaderboard split mathematics exist as a development module with reported settlement tests. They are not yet proven on a shared-world → entry → outcome path for a second sport.
- Green Goblin exemption, Red Demon cushion, offered-sides-only, non-padding, unique-player and same-event caps remain production law.

2.2 What is not solved

- No executable football PrimitiveStatRegistry against the frozen ledger.
- No conservation harness that rejects an impossible football world (sack double-count, receptions > targets, team pass yards ≠ sum of legal receiving yards under the declared definition).
- WorldProjectionResult is specified; the production-quality adapter that consumes it with a frozen EntryContract and emits WorldLineupOutcome on every world is not accepted as complete.
- CFB Reboot eligibility is explicitly not a copy of NFL. Without a frozen player registry it is UNRESOLVED.
- MLB plate-appearance / base-out state is designed in the blueprint and not implemented.
- Block-length selection is a frozen policy object with no coverage-tournament winner. Politis-White L^* is not BEL-optimal by theorem.
- Tracking feeds (Next Gen Stats, Second Spectrum, Statcast, Hawk-Eye, ball-tracking cricket) are not MarketDefinitions. Treating exit velocity or route separation as a prop primitive would contaminate the ledger.

- Leaderboard EV remains UNMODELED without a group-score distribution. That does not change in this ADR.

3. Proposed design

3.1 Two implement-now artifacts, one frozen plugin law

This ADR authorizes exactly two executable workstreams and one contract-level preparation layer.

Workstream A — Football Primitive Registry (NFL first). A sport plugin that fills EventLatentState, DiscreteEventRegime, team opportunity pools, player OpportunityState / EfficiencyState, PrimitiveStatLedger, ConservationRule set, and MarketDefinition bindings. No new dataclass fields.

Workstream B — E2E settlement path. A deterministic function chain: WorldProjectionResult × EntryContract × PlatformRuleVersion → RebootRuleRegistry → WorldPickState → LineupSettlement / WorldLineupOutcome → max(R_LB, R_MG) under payout_display_hash. First fixture: basketball worlds. Second fixture: NFL worlds. Same adapter.

Workstream C — Multi-sport appearance contract (design now, code stubs only if they compile against frozen schema). Every later sport registers an AppearanceModel: the discrete event atom that creates or withholds opportunity. Baseball plate appearances are the reference design. Football snaps/plays/targets, basketball possessions/stints, soccer minutes/touches, tennis points, golf holes/strokes, cricket balls, UFC fight-seconds follow the same object, not new schemas.

3.2 Causal stack (unchanged, sport-parameterized)

Sport → Event environment → Discrete regime → Team/side opportunity pool → Role → Player opportunity → Conditional efficiency → Primitive statistics → Derived statistics → MarketDefinition → Comparison against frozen line/side → Administrative translation → Economic translation.

Higher/Lower is never fit as a standalone classifier on “metrics.” Metrics enter as evidence that shrinks parameters or updates a regime posterior. The offered side is graded from the world distribution of the versioned market, not from a tracking highlight.

3.3 Why this layer is correct

- Football is the next sport in the frozen build order because its conservation surface is hostile: sacks, scrambles, targets vs receptions, dual-threat players, special teams zero-opportunity, defensive exclusion from many PrizePicks Reboot paths.
- If football fits the existing ledger/world/settlement objects, later sports are plugins. If football requires a new common field, the schema was not universal and must be argued as an impossibility — not quietly extended.
- E2E settlement is the load-bearing wall of v6 Phase C. Without it, card construction still optimizes a fictional 6-leg table.
- Appearance mechanics belong in the sport plugin, not in MarketDefinition and not in the optimizer. A plate appearance is a physical opportunity atom. A PrizePicks “batter ≤2 PA Reboot” rule is a platform translation of that atom.
- Block-length selection belongs in Phase A measurement, not in the football registry. Mixing them would smuggle a research selector into a production sport plugin.

4. Alternatives rejected

Alternative	Why rejected
Implement every requested sport in this patch	Violates build order, explodes conservation risk, and pretends unverified platform rules are known. Empty card is legal; fabricated edge is not.
New common schema fields per sport (pitch_id, hole_id, rally_id as first-class schema types)	Football already showed a different event structure can live inside EventLatentState payloads. New common types require an impossibility proof.
Fit Higher/Lower directly on Statcast / NGS / SportVU features	Collapses evidence into probability, mixes units, ignores conservation, and cannot settle. Tracking is Q-vector input, not a ledger primitive.
Copy NFL Reboot rules onto CFB, AFL, CFL, NFL preseason	Snapshot V1 forbids this. CFB needs an explicit player registry. Preseason is excluded. Other leagues are unknown platforms until verified.
Treat WAR, Strokes Gained, or IBM Watson-style outputs as primitives	Those are derived, often proprietary, often restated, and are not PrizePicks markets. They may enter evidence or research challengers only.
One generic Normal box-score sampler per player-prop	Reopens the exact failure v6 exists to close: PRA independently sampled from PTS/REB/AST.
Infer payout from card size or published 6-pick examples	Forbidden by EntryContract. displayed payout is captured at submission.
Collapse DNP / Reboot / Tie into void	Breaks payout-tier vs eligibility-population accounting and same-team refund logic.
Promote BEL_FIXED_B or Bartlett because football is next	Phase A roles are frozen. Coverage harness has not granted production authority.
Redesign basketball while adding football	Canonical-adjacent basketball layer is LIVE. Rewrite risk with no predictive mandate.
Claim LR000001 or predictive superiority at patch complete	Software version ≠ learning revision. No eligible future settlements have been earned.

5. Frozen invariants this ADR will not touch

- Canonical v5.4.1 installation bytes and source hash
bd1fb433d5f82d3812e453c30edcbb67db11b20f60e43cf50424c45a7c2ff474.
- Schema freeze PHASE_BC_SCHEMA_V1_2026-08-25 and contract hash
6e78dacc19843338643bdcabc7477fd3ce2dd065da1e9629646dacc21cdb1f22.
- Hash lineage with no skipped stages and no silent replacement of populated hashes.
- Derived values never overwrite primitives.
- Unknown state is explicit. Never coerce to WIN, LOSS, 0, false, or eligible.
- GOBLIN: analytics and historical settlement allowed; selection forbidden on every path including debug and portfolio repair.
- Red Demon extra cushion remains demotion-only.
- Offered sides only. Six legs are a target, never a fill requirement.
- Primitive validity ≠ PrizePicks selectability.
- Sports truth ≠ platform truth. Post-settlement official revisions are SPORT_STAT_REVISION_AFTER_SETTLEMENT.
- Leaderboard return UNMODELED unless a group distribution is actually available.
- Bartlett, DPMM, GP-LVM authority unchanged.

6. Schema impact

VERDICT: ZERO COMMON-SCHEMA CHANGE

A new sport requiring different primitives is not sufficient justification to alter PHASE_BC_SCHEMA_V1_2026-08-25.

Football payloads live in EventLatentState envelopes, sport-keyed ConservationRule rows, and sport-keyed PrimitiveStatRegistry configs.

Appearance atoms (PA, snap, possession, ball, point, hole, fight-second) are values of existing regime / opportunity objects, not new schema classes.

If implementation discovers a true impossibility, stop and produce the seven-part mutation dossier required by NEXT-DESIGN §3. Do not “just add a field.”

Post-patch check: `sha256(schemas/Phase_BC_Immutable_Contracts.json) == 6e78dacc19843338643bdcabc7477fd3ce2dd065da1e9629646dacc21cdb1f22`.

7. Workstream A — Football primitive registry

7.1 Event latents (NFL)

EventLatentState payload (sport-specific, not new schema fields): pace/play-rate, scoring environment, competitiveness / game script, weather, venue / surface, officiating tendency, implied total / spread as market-evidence not as outcome, discrete injury and starting-QB regime, rest / short-week flag, dome-vs-outdoor.

DiscreteEventRegime examples: STARTER_QB_ACTIVE / QUESTIONABLE_LIMITED / OUT; WR1_ACTIVE / OUT; NORMAL_SCRIPT / BLOWOUT_SCRIPT; ROOF_OPEN / CLOSED; PRECIP_NONE / RAIN / SNOW. Worlds draw regimes from the frozen pregame posterior. They do not convert uncertainty into one arbitrary minute penalty.

7.2 Team opportunity pools

The team layer owns the exhaustively allocated pools. Players receive shares of those pools. Efficiency is applied after opportunity.

Pool	Unit	Allocation notes	Do not confuse with
Offensive plays	count	Pass attempts + rush attempts + sacks taken, under declared sack semantics	Sum of all player snaps
Dropbacks	count	Attempts + sacks + scrambles (declared definition)	Designed pass plays only
Designed rushes	count	Team designed run pool	QB scrambles
Targets / routes	count	Dirichlet or multinomial share among eligible receivers	Receptions
Red-zone plays / goal-to-go	count	Optional path refinement when TD markets need ordering	Fantasy points
Special-teams opportunities	count	K/P only; zero for most skill players	Offensive snaps
Defensive opportunities	count	Physical model may include them	PrizePicks Reboot eligibility — defense excluded under V1

Hard non-identity: sum of player offensive snaps / team offensive plays. Multiple players occupy the same play. Snap totals are occupancy measures, not a conserved play-count.

7.3 Player opportunity vs efficiency

Role family	OpportunityState examples	EfficiencyState examples
QB	Dropbacks, designed rushes, scramble rate context, red-zone dropbacks	Completion rate, YPA, sack rate, INT rate, scramble YPC, TD rate per relevant attempt
RB	Designed carries, routes, targets, goal-line carries	YPC, catch rate, YPR, TD rate
WR/TE	Routes, targets, red-zone targets, air-yard share if evidenced	Catch rate, YPR / YPT, TD rate
OL	Pass-block / run-block snaps if modeled	Not a PrizePicks skill-player market in V1

Role family	OpportunityState examples	EfficiencyState examples
K / P	Attempt pool can be zero under specified game paths	Make rate / net punt — only if a verified market exists
Defense / IDP	Team opponent dropbacks, rush attempts	Selectable only if a verified MarketDefinition exists; Reboot path excluded under V1

7.4 Primitive topology (NFL)

Family	Keys (illustrative, registry-owned)	Semantic type
Volume / opportunity	off_snaps, routes, targets, dropbacks, pass_att, designed_rush_att, scramble_att, rush_att, rec_targets, rz_att	PRIMITIVE
Passing outcomes	pass_cmp, pass_yds, pass_td, int, sacks_taken, sack_yds, scramble_yds	PRIMITIVE
Rushing outcomes	rush_yds, rush_td, fumbles_lost (if in definition)	PRIMITIVE
Receiving outcomes	receptions, rec_yds, rec_td	PRIMITIVE
Kicking	fg_att, fg_made, xp_att, xp_made, punt_att (only if modeled)	PRIMITIVE
Derived / combo	pass+rush yds, rush+rec yds, rec+rush TD, anytime TD, fantasy / platform score	DERIVED / COMPOSITE / PLATFORM_SCORE

No simulator method writes a derived combo into the primitive ledger. pass+rush yards is f(pass_yds, rush_yds; definition_version).

7.5 Conservation and bound rules

Rule type	Expression (declared semantics)	Failure code
COMPONENT_IDENTITY	rush_att = designed_rush_att + scramble_att	PRIMITIVE_CONSERVATION_FAILURE
COMPONENT_IDENTITY	dropbacks = pass_att + sacks_taken + scramble_att	PRIMITIVE_CONSERVATION_FAILURE
SUM_EQUALS	team_off_plays = pass_att + rush_att + sacks_taken	PRIMITIVE_CONSERVATION_FAILURE
BOUND	pass_cmp ≤ pass_att; receptions ≤ targets; targets ≤ routes when routes exist	PRIMITIVE_CONSERVATION_FAILURE
RESOURCE_ALLOCATION	Σ player targets = team legal targets (eligible receivers only)	PRIMITIVE_CONSERVATION_FAILURE
SUM_EQUALS	team rec_yds reconcile with team pass_yds under sack/scramble/lateral definition	PRIMITIVE_CONSERVATION_FAILURE
LOGICAL	pass_td + rush_td + rec_td accounting does not double-count a single scoring play at team level	PRIMITIVE_CONSERVATION_FAILURE
BOUND	K/P attempt pools may be zero; skill-player special-teams attempts default zero unless evidenced	ok / explicit zero
LOGICAL	made ≤ attempts on every attempt family	PRIMITIVE_CONSERVATION_FAILURE

Phase B acceptance for football: P(structural invariant failure) = 0 on deterministic identities. Generate worlds that cannot violate; valid=False is defensive audit only.

7.6 Path vs aggregate

Default: aggregate team-pool + share + efficiency is enough for most full-game passing yards, rushing yards, receptions, and combo markets. Path model (drive / play / down-distance / score differential) is required when ordering changes the market: anytime TD, first-TD, completions on a drive-constrained script, blowout sit, weather-delayed second-half rest. Hybrid is allowed. Full play-by-play for every prop is not required and is not this patch's runtime budget.

7.7 CFB, AFL, CFL, NFL preseason — same plugin, different authority

League	Physical plugin	Platform / Reboot authority
NFL regular + postseason	IMPLEMENT NOW	Snapshot V1: MORE, full-game, offense eligible, K/P zero-opportunity path, defense excluded, no preseason.
NFL preseason (NFLP)	Physical worlds RESEARCH / ANALYTICS_ONLY	FAIL CLOSED on Reboot and settlement inference. Do not copy regular-season rows.

League	Physical plugin	Platform / Reboot authority
CFB	Registry may be drafted after NFL harness is green	IMPLEMENT settlement only with an explicit frozen player_reboot_eligible list. No Playoff-language shortcut.
CFL / AFL / UFL / other football	DEFER	Different play definitions (3-down, roster, motion). Unknown PrizePicks rows → UNKNOWN_PLATFORM_RULE.

8. Workstream B — E2E settlement path

8.1 Function contract

settle_world_lineup(world_projection, entry_contract, platform_rule_version, reboot_registry, participation_registry) → WorldLineupOutcome

Deterministic. Hash-replayable. No I/O. No nearest-table match. Missing display hash or pick field → ENTRY_CONTRACT_INCOMPLETE. Missing rule row → UNKNOWN_PLATFORM_RULE / UNKNOWN_REBOOT_RULE / UNKNOWN_PARTICIPATION_RULE. Authoritative comparison unavailable → PLATFORM_SETTLEMENT_UNRESOLVED.

8.2 Per-pick state machine

Order is mandatory: administrative state first, comparison only if ACTIVE, economic translation last.

Dimension	States	Job
AdministrativeState	ACTIVE, DNP, REBOOT, CANCELLED, INVALID_MARKET, UNRESOLVED	Was the pick removed or rewritten by platform administration?
ComparisonState	WIN, LOSS, TIE, NOT_APPLICABLE	Did the official / simulated stat beat, miss, or push the frozen line?
EconomicState	COUNTS_AS_WIN, COUNTS_AS_LOSS, TIER_REDUCTION, REMOVED, UNRESOLVED	How does this pick enter payout-tier and eligibility math?

LESS is not Rebooted under snapshot V1. If a MORE projection is already achieved before a qualifying early exit, settle normally — do not Reboot a won pick. Unknown eligibility is UNRESOLVED, not guessed TRUE.

8.3 Two card counts and lineup accounting

- original_pick_count is frozen from the contract.
- administrative_removed_count = DNP + Reboot + analogous removals.
- tie_count is separate. Ties step payout tier but remain in eligibility population.
- payout_tier_count ≠ eligibility_population_count.
- distinct_remaining_team_count evaluated after administrative removals. Remaining nonempty same-team set → refund economic state, not a rewritten comparison on survivors.
- Identities: active + void/DNP + reboot = contract pick count; wins + losses + pushes = active picks.
- R_final = max(R_Leaderboard, R_MinimumGuarantee) under frozen display. They do not stack.
- P(payout>0), P(net>0), E[gross], E[net], CVaR are different metrics. Report them separately.
- Leaderboard points: Demon 1.05, Standard 1.00, Goblin/Discount/Stack 0.95, same-player stack win 0.95 override. Loss/tie/DNP/Reboot/unresolved = 0. Split L among tied top scorers only.
- expected_leaderboard_return_iid remains RESEARCH_ONLY.

8.4 First fixtures

Fixture 1: synthetic basketball EventWorld already passing conservation → MarketDefinition projections → 2-pick and 6-pick EntryContracts covering WIN/LOSS/TIE, DNP, NBA 1H-leave Reboot, same-team refund, MG vs Leaderboard max, GOBLIN present in analytics contract but rejected if selection path attempted.

Fixture 2: synthetic NFL EventWorld → passing/rushing/receiving projections including a combo → same adapter with NFL Reboot rows, K/P zero-opportunity, defense exclusion, unknown stat key UNRESOLVED.

Fixture 3: intentional corruption of each ledger must die before settlement with PRIMITIVE_CONSERVATION_FAILURE or DERIVED_IDENTITY_FAILURE. Settlement code must never “fix” a broken world.

9. Multi-sport readiness — plugin map, not a build-all mandate

Every sport below can produce a Higher/Lower conclusion if and only if four things exist: (1) an appearance/opportunity atom, (2) an efficiency model, (3) a conservation set with $P(\text{failure})=0$ on identities, (4) a verified MarketDefinition + participation/reboot row. Missing any one of those four is FAIL CLOSED for selection, even if the physical simulator can still run for research.

Sport / league	Appearance atom	Primary opportunity pools	Status this ADR
NBA	Possession / stint	Minutes, FGA, 3PA, FTA	LIVE primitives. Reuse through E2E path.
WNBA	Possession / stint	Minutes, FGA, 3PA, FTA	LIVE primitives. Reboot analogous; do not assume NBA stat-key list.
NFL	Play / drive / snap occupancy	Plays, dropbacks, rushes, targets, routes	IMPLEMENT NOW.
NFL preseason	Same physical atom	Reduced / unstable role pools	ANALYTICS_ONLY. Settlement FAIL CLOSED.
CFB	Play / drive / snap	Same family as NFL, different roster entropy	After NFL harness. Reboot requires player registry.
CFL	Play (3-down)	Plays, pass, rush; motion/roster differ	DEFER. Unknown platform rows.
AFL (Aussie rules)	Disposal / contest / time-on-ground	Disposals, marks, tackles, inside-50	DEFER. New identities; same schema envelope.
MLB	PA / base-out / inning	PA, BF, pitches, outs	Design in this ADR. Next registry after football E2E.
MiLB / NPB / KBO / college baseball	PA / BF	Same atoms, different run environments	DEFER until MLB plugin exists.
Soccer / EPL / MLS / UCL	Minutes / on-ball events / set pieces	Minutes, shots, touches, corners, saves	DEFER. Low-score discrete + heavy zero inflation.
Tennis	Point / game / set / match	Serve points, return points, break chances	DEFER. Path model almost mandatory.
Golf	Stroke / hole / round	Holes started, putts, SG components as DERIVED	DEFER. Weather + course as event latents.
Cricket	Ball / over / innings	Balls faced, balls bowled, overs	DEFER. Baseball-like depth; different identities.
UFC / MMA	Round / fight-second	Fight seconds, strike attempts, TD attempts	PLANNED in blueprint. After baseball or in parallel research.
Lacrosse (LAX)	Possession / minutes	Possessions, shots, faceoffs	DEFER.
Handball	Possession / minutes / attack	Attacks, shots, goalkeeper minutes	DEFER.
OTD / unnamed board	Unknown	Unknown	FAIL CLOSED until defined and verified.

9.1 Tracking data law

Statcast (exit velocity, launch angle, barrel), NBA spatial tracking (25 Hz coordinates), NFL Next Gen Stats (RFID speed, separation, route), Hawk-Eye / tennis shot-spotting, cricket ball-tracking, golf launch monitors are evidence, not primitives.

Allowed: they update EfficiencyState posteriors or DiscreteEventRegime probabilities through the EvidenceGraph with as-of timestamps. Forbidden: writing exit_velocity into PrimitiveStatLedger as if it were PrizePicks “Hits”; generating a Higher/Lower from a tracking residual without a MarketDefinition; treating WAR or Strokes Gained as a conserved primitive. Those metrics can support research challengers and evidence quality vectors. They cannot skip the world.

9.2 How any sport reaches Higher / Lower

For offered side $s \in \{\text{MORE}, \text{LESS}\}$ on market m with line L and definition version v :

1. Freeze evidence before cutoff. 2. Draw EventWorlds from the sport plugin. 3. Write only primitives into the ledger. 4. Enforce conservation. 5. Compute $x_w = f(\text{ledger}_w; v)$. 6. ComparisonState_w from $(x_w, L, s, \text{push policy})$. 7. $\hat{p}$ = mean of world indicators after administrative filters that are knowable pregame; in-play admin remains UNRESOLVED when not simulable. 8. Grade PLAYABLE / LEAN / PASS / TRAP using probability, LCB, reliability, fragility, Demon cushion — never using a single collapsed confidence number.

No sport gets a shortcut that fits a logistic model on “advanced metrics” and stamps PLAYABLE. That would recreate independent-composite sampling with better vocabulary.

10. Appearance mechanics

An appearance is the discrete event that creates a unit of opportunity for a participant. It is the missing language between “minutes” and “box score.” Different sports use different atoms; the object is always OpportunityState generated by an AppearanceProcess.

10.1 Universal AppearanceProcess

AppearanceProcess_v1 (payload inside existing opportunity / regime objects):

- atom_type $\in \{\text{POSSESSION}, \text{STINT}, \text{PLAY}, \text{SNAP_OCCUPANCY}, \text{DRIVE}, \text{PA}, \text{BF}, \text{PITCH}, \text{BALL}, \text{OVER}, \text{POINT}, \text{GAME_SET}, \text{HOLE}, \text{STROKE}, \text{ROUND}, \text{FIGHT_SECOND}, \text{MINUTE}, \text{TOUCH}, \text{SHOT}, \text{FACEOFF}, \text{ATTACK}\}$
- eligibility_rule — who can receive the next atom (lineup, batting order, on-field XI, serve holder, QB in game).
- generation_process — how the next atom arrives (Markov base-out, drive transition, possession exchange, scheduled hole, timed minutes).
- termination_rule — when the atom stream stops (game clock, three outs, 27 outs, fight finish, retirement, red card, DNP).
- zero_opportunity_paths — explicit states that emit zero atoms (scratched pitcher, K/P with no attempt, player sat in blowout, batter does not reach spot in order).
- platform_participation_map — how atom counts translate to DNP / Reboot / min-PA / min-minutes. This map is versioned with the platform, not inferred from physics.

This is why MLB “ ≤ 2 PA Reboot” is not a baseball identity. It is a platform map over a baseball appearance process. Physics can produce 1 PA. Settlement then asks whether that 1 PA meets the frozen Reboot row.

10.2 Baseball plate-appearance mechanics (design-now)

MLB is the most state-dependent of the requested sports. Implementing it before football E2E would violate build order. Designing it now prevents the football plugin from painting the schema into a snap-only corner.

Physical generation

- The game is a sequence of plate appearances embedded in a base-out-inning state $S = (\text{inning}, \text{half}, \text{outs}, \text{bases}, \text{score}, \text{pitcher_id}, \text{batter_id}, \text{lineup_slot})$.

- A plate appearance is one batter’s opportunity atom until the PA terminates: hit, out, walk, HBP, sacrifice, catcher interference, etc. Balls in play that become fielder’s choices still consume a PA.
- Identity: $PA = AB + BB + HBP + SF + SH + CI$ (+ any other official non-AB PA codes the definition version names).
- Identity: $H = 1B + 2B + 3B + HR$. $TB = 1 \cdot 1B + 2 \cdot 2B + 3 \cdot 3B + 4 \cdot HR$.
- Bound: $AB \geq H$; $K \leq PA$; $HR \leq H$; RBI cannot exceed runs generated by that PA under scoring rules; pitcher BF is the complementary batter-facing atom.
- Opportunity and efficiency stay split: PA/BF/pitches are opportunity; K%, BB%, EV/LA residual, wOBA-on-contact are efficiency. Do not “fix” a short-PA role with a huge EV prior.
- Starting pitcher vs reliever vs opener vs bulk are DiscreteEventRegime values. Historical PAs from another role epoch are not exchangeable without shrinkage.
- Path vs aggregate: full-game pitcher strikeouts and batter H+R+RBI usually need base-out path or a carefully calibrated aggregate that still conserves team runs. Stolen bases, RBI, and runs are path-heavy. Isolated pitcher K can often be $BF \times K\text{-rate}$ with BF itself coming from a pitch-count / lineup-turn model.

Platform map (snapshot V1, already frozen)

- MLB batter MORE full-game; Reboot if batter leaves with ≤ 2 PA. Pitchers excluded from that Reboot path. MLBLIVE excluded. PA, not AB.
- Unknown batter-stat eligibility → UNRESOLVED. Do not infer from another book’s “hits” market.

Implementation of the MLB registry is not authorized by this ADR’s IMPLEMENT NOW class. The appearance contract above is authorized as the design target the football plugin must not preclude.

10.3 Appearance atoms for the rest of the requested set

Sport	Atom	What “did not appear” means	Typical zero-opportunity paths
NBA / WNBA	Stint / possession while on floor	DNP-CD or injury scratch; minutes = 0	Out; G-League; load-manage; foul-out late is not zero, it is truncated
NFL / CFB	Play participation / target / carry	Inactive or in game with zero designed opportunities	Inactive; emergency 3rd QB; K/P with no attempt; blowout sit
Soccer	Minute on pitch + set-piece role	Not in XVIII or not introduced	Unused sub; red card after minute t truncates remaining minutes
Tennis	Point on serve or return	Retirement / walkover	Walkover is not a 0-0 tennis world; it is an administrative outcome
Golf	Hole started	WD / DQ / cut (cut is a mid-event termination)	Missed cut truncates rounds 3–4; do not impute Saturday scores
Cricket	Ball faced or ball bowled	DNB (did not bat) vs 0 off N balls	DNB ≠ 0 runs. That distinction is an identity, not a style choice
UFC	Fight-second in a state	Replacement / weigh-in miss / finish ends time	KO in round 1 zeros later-round striking totals
LAX / handball	Possession / minute / shot clock attack	DNP / unused	Specialist roles may have near-zero shot opportunity

Cricket DNB versus 0 off 12 balls is the cleanest proof that appearance must be first-class. Treating DNB as a zero would poison both the physical model and any Reboot analogue.

11. Block-length selection criteria

Block-length selection is not part of the football code path. It is Phase A measurement policy. It is documented here because the original mission and the user request both require it, and because it must not be “solved” inside a sport registry.

11.1 What is already frozen

- No universal optimum exists between long-run variance estimation and BEL coverage.
- Politis-White (2009-corrected) is a candidate generator for a center L^* , not a theorem that L^* is optimal for BEL coverage.
- Small-b BEL and fixed-b BEL address different approximation errors. Bartlett does not replace fixed-b and does not solve convex-hull failure.
- Method roles stay frozen: BEL_CHI2 diagnostic; BEL_BARTLETT research-only; BEL_FIXED_B main temporal robustness challenger; ABEL convex-hull robustness; PBEL block-choice robustness challenger.
- Do not use the i.i.d. scalar EL Bartlett factor on overlapping BEL.

11.2 BlockPlan fields that must be populated

block_selection_objective, block_selector_version, block_selector_training_hash, plus the already-specified plan fields: time_unit, sport_family, market_family, training_cutoff, regime_id, regime_break_rule, block_length_rule, block_lengths_evaluated, selected_block_length, overlap_rule, block_start_spacing, fixed_b_ratio, critical_value_version, plan_hash.

11.3 Hard invariants

- Enough valid blocks: $M \geq 1$, $L \geq 1$, and $Q = \text{floor}((N-M)/L)+1$ (or overlapping analogue) must be adequate for the method.
- $\text{training_cutoff} < \text{first_evaluation_timestamp}$. No post-outcome block shopping.
- No block crosses regime_id.
- $\text{time_unit} \in \{\text{SLATE}, \text{DAY}, \text{EVENT_WINDOW}\}$.
- Selection criterion is preregistered on training-only data.

11.4 Recommended engineering objective (not a theorem)

On the frozen training window, evaluate a small grid around a Politis-White center: $\{0.5, 0.75, 1.0, 1.5, 2.0\} \times L^*_{PW}$, subject to minimum valid blocks and regime boundaries. For fixed-b, evaluate a frozen grid of $b = L/n \in (0,1]$ and store critical values by b.

$J(L) = w_C |\text{Coverage}(L) - (1-\alpha)| + w_T |\text{TypeI}(L) - \alpha| + w_W \text{normalized interval width} + w_F \text{convex-hull / solver failure rate}$.

Initial engineering weights (priors only, not learned): emphasize Type-I and coverage over interval width. A narrow interval that undercovers is a promotion hazard. PBEL remains in the tournament because it avoids committing to one L.

Coverage harness stress: slate counts 20/40/80/160/320; serial $p \in \{0, 0.3, 0.6, 0.8\}$; Gaussian / skewed / heavy-tail / contaminated; balanced vs unequal clusters; stable / drift / break; null / weak / true superiority / true degradation. No method receives production promotion from this ADR.

11.5 Classification

RESEARCH FIRST. Football implementation must consume whatever BlockPlan is already frozen for measurement of later football forecasts. It must not choose L from football residuals after seeing football outcomes.

12. Data requirements

Need	For workstream	Available?	Gap action
Verified canonical v5.4.1 source + ledger hashes	All	Must be loaded before code	STOP if hash mismatch
Frozen schema JSON + introspection test	All	Specified	Must remain byte-identical
Basketball primitive + conservation tests	E2E fixture 1	Accepted LIVE	Reuse
PrizePicks snapshot V1 tables + 13-test module	E2E	Reported as development baseline	Reuse; do not rewrite math
NFL team play / dropback / target accounting definitions	Football registry	Must be declared in registry version notes	Pick one semantics and freeze it
NFL Reboot stat-eligibility list	E2E NFL fixture	Public page incomplete	VERIFIED_TRUE/FALSE registry; else UNRESOLVED
CFB qualifying-player list	CFB only	Ambiguous publicly	Do not ship CFB settlement without it
NFL preseason official PrizePicks rows	NFLP	Not in V1	FAIL CLOSED
Play-by-play for path markets	Football path subset	Nice, not required for v1 aggregate markets	Aggregate first
NGS / Statcast / SportVU feeds	Evidence / research	Not required for IMPLEMENT NOW	Do not block football on tracking
Group Leaderboard opponent scores	Platform EV	Unavailable	Keep UNMODELED
OTD definition	Unknown board	Not specified	FAIL CLOSED

13. Primary risks

Risk	Why it matters	Mitigation
False conservation (Σ snaps = plays)	Silently impossible worlds	Only enforce physically true identities; document scope and overtime
Sack / scramble double count	Pass yards and rush yards both claim the play	Declare one dropback identity in the registry version
Team pass yards $\neq$ receiver yards under laterals / sacks / scrambles	Composite markets contradict	Definition-versioned reconciliation rule; fail closed if unspecified
Copying NFL Reboot onto CFB or preseason	Wrong money	Separate registry keys; unknown $\rightarrow$ UNRESOLVED
Wiring settlement only to football	Adapter becomes sport-shaped	Basketball fixture first
Tracking-feature leakage into primitives	Unsettled, non-conserved "edge"	EvidenceGraph only
Choosing block length after seeing football results	Temporal firewall breach	BlockPlan frozen on training cutoff; football does not own L
Declaring TOTAL_PLATFORM_EV modeled	Leaderboard group unknown	Report MG modeled, LB unmodeled, total partial
Scope explosion across 15 sports	Nothing ships, schema mutates	This ADR authorizes two workstreams only
HAR fixture FileNotFound	False regression narrative	Keep pre-existing external gap separate from new tests

14. Validation plan

14.1 Tests required before the patch is accepted

- Unit: football identity formulas, bound checks, role zero-opportunity paths, K/P zero pool.
- Identity: every exact conservation row on valid synthetic games, including overtime if the definition includes it.
- Corruption: flip 2PA-style identities ($\text{cmp} > \text{att}$, $\text{rec} > \text{tgt}$, $\text{plays} \neq \text{pass} + \text{rush} + \text{sacks}$) and assert hard failure before projection.

- Market integration: ledger → MarketDefinition → WorldProjectionResult for pass yds, rush yds, rec, rec yds, pass+rush, rush+rec, anytime TD if definition verified.
- Settlement integration: basketball E2E + NFL E2E covering DNP, Reboot, Tie, same-team refund, MG, Leaderboard split, unknown rule, Goblin rejected from selection constructor.
- Accounting: the two lineup identities on every fixture.
- Hash: lineage complete; no mutation after content_hash; schema file hash unchanged.
- Temporal: no future evidence in frozen worlds.
- Regression: previously accepted basketball / Phase B-C / v5.x runnable tests remain green. External HAR misses stay skipped/environment.
- Fresh-baseline reapplication: clean copy of accepted development baseline + this patch only.

14.2 What this patch does not validate

It does not validate predictive CRPS/ES/VS improvement. It does not promote a Learning Revision. It does not calibrate $P(K \geq 5)$. It does not grant BEL_FIXED_B production authority. It does not make Leaderboard EV modeled. It does not authorize AFL/CFL/LAX/handball/soccer/tennis/golf/cricket production selection.

14.3 Promotion gates if football worlds later become a forecasting challenger

Future-only primary Energy Score or predeclared CRPS on the relevant market family; companion noninferiority; clustered WCR-T; at least one temporal method noninferior; no validated harm; subgroup safety; slate stability; conservation still $P(\text{failure})=0$. Until then PRODUCTION_AUTHORITY on probabilities remains the inherited v5.4.1 basketball-era machinery, not “the new football sampler is live therefore it is better.”

15. Production state at launch of this patch

Software label	Forward-development patch on accepted baseline. Not a silent v7. Not LR000001.
Football registry	DEVELOPMENT BASELINE, executable, conservation-tested
E2E path	DEVELOPMENT BASELINE, executable on basketball + NFL synthetic/real frozen contracts
Predictive superiority	NONE
Chronological calibration	INACTIVE / INSUFFICIENT ELIGIBLE SETTLEMENTS
Card-tail calibration	UNPROVEN
CFB settlement	LOCKED unless player_reboot_eligible registry present
NFL preseason settlement	LOCKED
Other sports	Schema-ready plugin slots only
Leaderboard EV	UNMODELED_UNLESS_GROUP_DISTRIBUTION_AVAILABLE
TOTAL_PLATFORM_EV	LOWER_BOUND_OR_PARTIAL
DPMM / GP-LVM / Bartlett	Unchanged research posture
Strict Card	FAIL_CLOSED on inherited gates; no padding

16. Implementation blueprint (authorized modules only)

Exact targets. Anything not listed is out of scope for the IMPLEMENT NOW patch.

Module	Responsibility	Must not do
sports/football/registry.py	Primitive keys, units, families, market bindings	Write derived combos into the ledger

Module	Responsibility	Must not do
sports/football/conservation.py	phase_bc_conservation_rules() football rows	Invent Σ snaps = plays
sports/football/opportunity.py	Team pools + player shares	Fold efficiency into usage
sports/football/efficiency.py	Role-conditional rates	Repair missing snaps with huge YPC
sports/football/appearance.py	Play/snap/target appearance process payload	New schema class
sports/football/ledger.py	build_phase_bc_primitive_ledger() football	Mutate after hash
platform/prizepicks/settlement.py	settle_world_lineup(...)	Nearest payout table; guess unknown Reboot
platform/prizepicks/reboot.py	Snapshot V1 lookups	Share NFL rows with CFB/NFLP
platform/prizepicks/entry_contract.py	Freeze display hash + picks	Reconstruct L from card size
selection/eligibility.py (hook only)	Reject Goblin / incomplete contract	Force six legs
validation/conservation_harness.py	Valid + corrupt worlds for NBA + NFL	Skip corruption tests
tests/test_football_registry.py	Identities, bounds, roles	Weaken tolerances
tests/test_e2e_world_to_lineup.py	Basketball then NFL fixtures	Sport-specific adapter forks
schemas/ Phase_BC_Immutable_Contracts.json	Unchanged	Any field add

17. Engineering verdict

Axis	Rating	Note
Expected engineering benefit	High	Unlocks the first non-basketball proof of the frozen schema and the first honest platform-return path.
Expected predictive benefit at launch	None claimed	A working ledger is not a better forecast.
Implementation risk	Medium	Sack/scramble/target identities are easy to get subtly wrong.
Overfitting risk of this slice	Low	No new probability model is being promoted.
Runtime cost	Acceptable	Aggregate NFL worlds first; path models only where ordering matters.
Architectural fit	Exact	Master Blueprint §12.1 steps 3-4 and §14.4.
Schema fit	Exact if discipline holds	Zero common-schema change.
Multi-sport request fit	Honest partial	Appearance contract + plugin map now; other leagues deferred rather than faked.
Should it be built?	YES — IMPLEMENT NOW	The two workstreams only. Everything else stays classified as in § Decision summary.

18. Self-audit against NEXT-DESIGN §41 and mission §49

Question	Answer
Force six picks?	No.
Goblins selectable?	No path into production EntryContract.
Remove Demon cushion?	No.
Evidence quality as probability?	No. Q-vector stays separate.
Mix opportunity and efficiency?	No. Separate objects in football plugin.
Independently sample composites?	Forbidden. Combos derived.
Forecast-dependent VS as proper score?	Not in this patch.
Mixed units without transform?	Not in this patch.
Energy $\beta=2$ as strictly proper?	Not touched.
Mix empirical vs U-stat estimands?	Not touched.
Ordinary EL under serial dependence?	Not touched. BEL roles frozen.

Question	Answer
BEL production-authoritative without coverage?	No.
i.i.d. Bartlett on overlapping BEL?	Explicitly forbidden.
Block length post-outcome?	Forbidden. Training cutoff invariant.
Any inferential disagreement = model loss?	No. Not in this patch.
n_clusters = n_eff?	No.
Infer payout from card size?	No. EntryContract display hash.
Guess unknown Reboot eligibility?	No. UNRESOLVED.
Collapse DNP/Reboot/Tie?	No.
Treat Leaderboard payout as known without group info?	No. UNMODELED.
Confuse payout>0 with net>0?	No. Separate metrics.
Later sport-stat corrections rewrite platform economics?	No.
DPMM clusters as truth?	Unchanged shadow rules.
Claim superiority at LR000000?	No.
Silently replace v5.4.1?	No.
Invent false conservation?	Snaps-sum explicitly rejected.
Change frozen schema?	No.
Implement 15 sports because they were listed?	No. Classified and deferred.
Use WAR / Strokes Gained / Watson as primitives?	Rejected.
Copy NFL rules onto NFL preseason / CFB / CFL / AFL?	Rejected.

19. Decision

IMPLEMENT NOW

1. NFL football PrimitiveStatRegistry + conservation harness + appearance/opportunity/efficiency split against PHASE_BC_SCHEMA_V1_2026-08-25.
2. Deterministic settle_world_lineup path: WorldProjectionResult → EntryContract → WorldLineupOutcome, fixture-proven on basketball worlds then NFL worlds.
3. Freeze football conservation semantics (dropback / sack / scramble / target) in a versioned registry note so later arguments cannot silently rewrite identities.

Do not implement the rest of the sport catalog in this patch. Do not mutate the common schema. Do not promote a Learning Revision. Do not call the result predictively superior.

RESEARCH FIRST / DEFER / FAIL CLOSED (binding)

RESEARCH FIRST: BlockPlan coverage tournament; MLB PA / base-out registry as the next sport plugin after this patch is green; path-level NFL drive models for TD-ordering markets.

DEFER: AFL, CFL, LAX, handball, soccer/EPL, tennis, golf, cricket, UFC production registries. Keep them as plugin slots under the AppearanceProcess contract.

FAIL CLOSED: NFL preseason settlement inference; CFB settlement without player_reboot_eligible; OTD / unnamed boards; any Higher/Lower generated from tracking metrics without a verified MarketDefinition.

End of ADR-V6-001. Canonical v5.4.1 untouched. Schema hash must not move. Next artifact after acceptance is implementation of Workstreams A and B only.